

WONJU, Korea, Republic of

WMO: 471140

Lat: 37.3375N

Lon: 127.9467E

Elev: 150

StdP: 99.54

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.3	-11.3	-22.3	0.5	-7.8	-20.0	0.6	-6.9	7.0	1.2	5.9	-1.3	1.0	50	0.378

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.9	33.0	24.6	31.6	23.6	30.2	22.9	26.1	30.0	25.3	29.1	24.7	28.2	2.2	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.1	20.5	27.7	24.2	19.5	27.1	23.7	18.8	26.8	82.1	29.8	78.7	28.8	75.8	28.6	29.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5.7	4.9	4.2	DB	-16.3	34.9	2.5	1.5	-18.0	35.9	-19.5	36.8	-20.9	37.6	-22.6	38.7
			WB	-16.9	26.8	2.3	1.3	-18.6	27.8	-19.9	28.5	-21.2	29.3	-22.9	30.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	12.2	-3.0	-0.1	5.5	12.2	17.9	22.5	25.5	25.8	20.6	13.9	6.3	-0.9	(6)
(7)		DBStd	10.60	4.10	4.02	3.73	3.68	2.69	2.23	2.20	2.56	2.80	3.50	4.26	4.13	(7)
(8)		HDD10.0	1326	404	284	146	19	0	0	0	0	9	125	339		(8)
(9)		HDD18.3	2913	662	517	397	186	41	2	0	11	139	361	597		(9)
(10)		CDD10.0	2144	0	1	7	86	246	374	479	490	317	132	14	0	(10)
(11)		CDD18.3	689	0	0	0	2	28	126	221	232	78	3	0	0	(11)
(12)		CDH23.3	5579	0	0	0	42	279	917	1819	2063	442	18	0	0	(12)
(13)		CDH26.7	1728	0	0	0	4	44	247	592	760	82	0	0	0	(13)
(14)	Wind	WSAvg	1.4	1.3	1.4	1.8	1.8	1.6	1.4	1.4	1.3	1.1	1.1	1.2	1.1	(14)
(15)	Precipitation	PrecAvg	1281	24	28	51	86	94	142	333	253	151	45	44	31	(15)
(16)		PrecMax	2068	76	85	112	211	226	428	615	719	583	166	92	79	(16)
(17)		PrecMin	900	4	0	15	12	21	27	78	37	4	2	7	4	(17)
(18)		PrecStd	252	20	24	27	54	51	101	149	153	134	36	25	17	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.2	14.2	20.2	27.0	29.8	32.1	34.9	35.0	30.8	25.2	19.8	12.0	(19)
(20)		0.4%	MCWB	5.3	9.0	11.2	15.3	17.4	20.3	25.3	24.8	22.5	17.1	13.4	7.5	(20)
(21)		2%	DB	6.5	11.1	17.4	24.3	27.9	30.5	32.8	33.4	28.9	23.4	17.1	9.1	(21)
(22)		2%	MCWB	2.8	5.4	9.3	14.0	16.5	20.1	24.8	24.8	21.2	16.4	11.9	5.5	(22)
(23)		5%	DB	4.8	8.7	15.0	22.2	26.3	29.3	31.2	32.1	27.3	21.8	15.3	7.2	(23)
(24)		5%	MCWB	1.6	4.0	8.0	12.6	16.1	19.8	23.9	24.4	20.2	15.4	10.7	4.2	(24)
(25)		10%	DB	3.1	6.4	12.9	20.1	24.8	28.1	29.9	30.7	26.0	20.1	13.4	5.4	(25)
(26)		10%	MCWB	0.4	2.3	6.8	11.5	15.5	19.5	23.6	24.1	19.7	14.7	9.6	2.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.7	10.8	13.0	18.5	21.0	24.1	27.2	27.2	25.0	19.7	14.9	8.4	(27)
(28)		0.4%	MCDWB	7.8	12.5	16.7	23.7	25.1	27.2	31.4	31.2	27.6	22.0	17.5	10.8	(28)
(29)		2%	WB	3.6	6.2	10.9	15.9	19.1	22.9	26.1	26.2	23.4	17.9	13.1	6.2	(29)
(30)		2%	MCDWB	5.7	10.2	15.0	20.9	24.0	26.5	29.9	30.4	26.6	21.3	16.0	8.3	(30)
(31)		5%	WB	2.1	4.2	9.0	14.1	18.0	22.0	25.3	25.6	22.1	16.6	11.6	4.6	(31)
(32)		5%	MCDWB	4.1	7.8	13.7	19.7	22.8	26.0	28.9	29.5	25.2	20.4	14.3	6.9	(32)
(33)		10%	WB	0.8	2.7	7.4	12.8	16.9	21.2	24.6	24.9	20.9	15.5	10.1	2.9	(33)
(34)		10%	MCDWB	2.8	6.0	12.1	18.2	22.3	25.5	28.0	28.8	24.5	19.0	12.9	5.1	(34)
(35)	Mean Daily Temperature Range		MDBR	9.6	10.5	11.1	12.7	11.9	10.0	7.5	7.9	9.5	11.4	10.4	9.4	(35)
(36)		5% DB	MCDBR	10.2	13.3	14.7	16.0	14.7	12.4	9.8	9.9	11.0	12.8	12.7	11.2	(36)
(37)		5% DB	MCWBR	7.1	8.4	8.2	7.5	5.8	4.5	3.7	3.5	4.5	6.3	7.8	7.9	(37)
(38)		5% WB	MCDBR	9.0	11.7	12.8	13.3	11.6	9.4	8.3	8.5	8.7	10.6	10.7	10.5	(38)
(39)		5% WB	MCWBR	7.0	8.2	8.2	7.6	6.1	4.6	4.2	4.0	4.5	6.1	7.5	8.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.409	0.463	0.552	0.589	0.597	0.658	0.625	0.554	0.478	0.465	0.443	0.404	(40)	
(41)		taud	1.995	1.896	1.742	1.712	1.735	1.668	1.800	1.982	2.130	2.033	2.022	2.039	(41)	
(42)		Ebn at Noon	730	739	710	715	719	675	694	736	773	731	689	704	(42)	
(43)		Edh at Noon	132	166	213	232	230	246	214	175	143	143	128	119	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.50	3.26	4.17	4.88	5.51	5.12	4.04	4.18	3.94	3.52	2.46	2.23	(44)	
(45)		RadStd	0.17	0.28	0.44	0.38	0.48	0.43	0.59	0.59	0.47	0.26	0.28	0.17	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (48 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+89	+49

Nomenclature: See separate page